

Unit 2

Addressing the Challenges in Fog Edge Computing

TOPICS

- The Networking Challenges
- The Management Challenges
- Miscellaneous Challenges

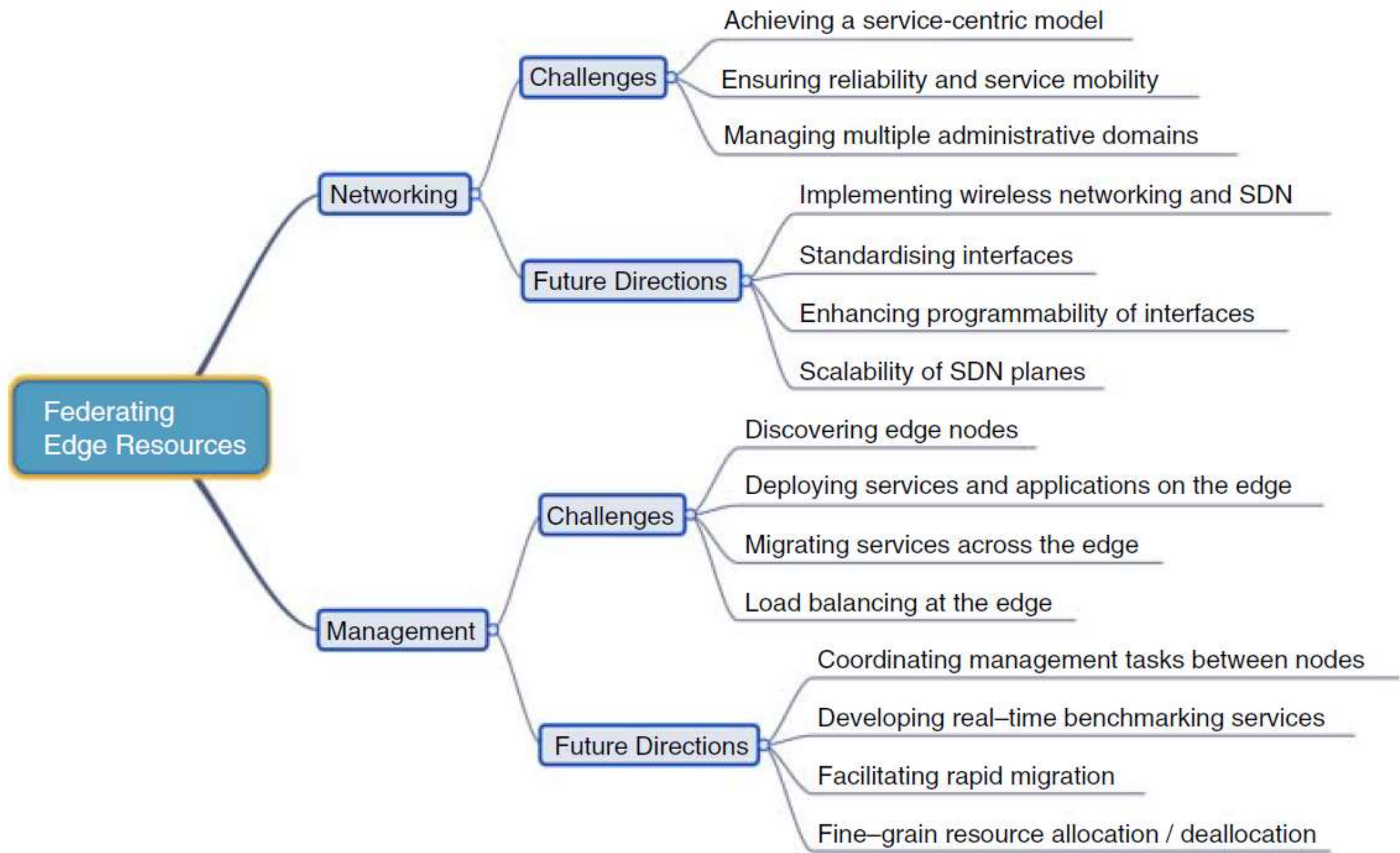


Figure 2.1 Networking and management challenges in federating edge resources.

I. Networking Challenges

1. Introduction to Networking Challenges

- **Definition of Edge Computing**

- Edge computing refers to the practice of processing data near the source of data generation rather than relying solely on centralized cloud data centers.
- It aims to reduce latency, improve bandwidth efficiency, and enhance the quality of service (QoS) for applications that require real-time processing.

- **Significance of Edge Computing**

- With billions of devices connected to the Internet, edge computing is essential for managing the increasing data traffic and ensuring responsive applications.
- It allows for localized data processing, which is crucial for applications like IoT, autonomous vehicles, and augmented reality.

Table 2.1 Network challenges, their causes and potential solutions in federating edge resources.

Networking challenge	Why does it occur?	What is required?
User mobility	Keeping track of different mobility patterns	Mechanisms for application layer handover
QoS in a dynamic environment	Latency-intolerant services, dynamic state of the network	Reactive behavior of the network
Achieving a service-centric model	Enormous number of services with replications	Network mechanisms focusing on “what” instead of “where”
Ensuring reliability and service mobility	Devices and nodes joining the network (or leaving)	Frequent topology update, monitoring the servers and services
Managing multiple administrative domains	Heterogeneity, separate internal operations and characteristics, different service providers	Logically centralized, physically distributed control plane, vendor independency, global synchronization

Networking Challenges

2. Key Networking Challenges

- **User Mobility**

- **Impact on Service Delivery**

- Users frequently move between different edge nodes, necessitating seamless service continuity.
 - The need for mechanisms that allow for application layer handover to maintain service quality.

- **Proposed Solutions**

- Implementing mechanisms that track user mobility patterns and facilitate smooth transitions between edge nodes.

Networking Challenges

- **Quality of Service (QoS) in a Dynamic Environment**
 - **Challenges**
 - Latency-intolerant services require consistent performance, which can be disrupted by network changes.
 - The dynamic state of the network necessitates a reactive approach to maintain QoS.
 - **Proposed Solutions**
 - Developing adaptive network protocols that can respond to changing conditions in real-time.

Networking Challenges

Achieving a Service-Centric Model

- **Transition from Host-Centric to Service-Centric**
 - Traditional models focus on the geographic location of servers, which limits flexibility.
 - A service-centric model emphasizes "what" service is needed rather than "where" it is located.
- **Challenges**
 - Coordinating service replication and ensuring unique service identifiers across multiple regions.
- **Proposed Solutions**
 - Implementing global service registries that allow for dynamic service discovery and access.

Networking Challenges

- **Reliability and Service Mobility**

- **Issues**

- Frequent connection and disconnection of devices can lead to an unreliable network environment.
 - Users expect seamless service access, which can be hindered by network instability.

- **Proposed Solutions**

- Service replication and migration strategies to ensure that services remain available even when nodes disconnect.

Networking Challenges

- **Managing Multiple Administrative Domains**

- **Challenges**

- Heterogeneity in network characteristics and operations across different administrative domains complicates management.
 - The need for a unified approach to manage services across diverse environments.

- **Proposed Solutions**

- Establishing a logically centralized control plane that can coordinate operations while allowing for distributed management within individual domains.

Networking Challenges

3. Proposed Solutions to Networking Challenges

- **Software-Defined Networking (SDN)**

- **Overview of SDN**

- SDN separates the control plane from the data plane, allowing for centralized management of network resources.
 - It provides a flexible and programmable network architecture that can adapt to changing conditions.

- **Role in Addressing Networking Challenges**

- SDN enables real-time monitoring and decision-making, which is crucial for managing dynamic edge environments.
 - It simplifies the management of user mobility and service delivery across multiple edge nodes.

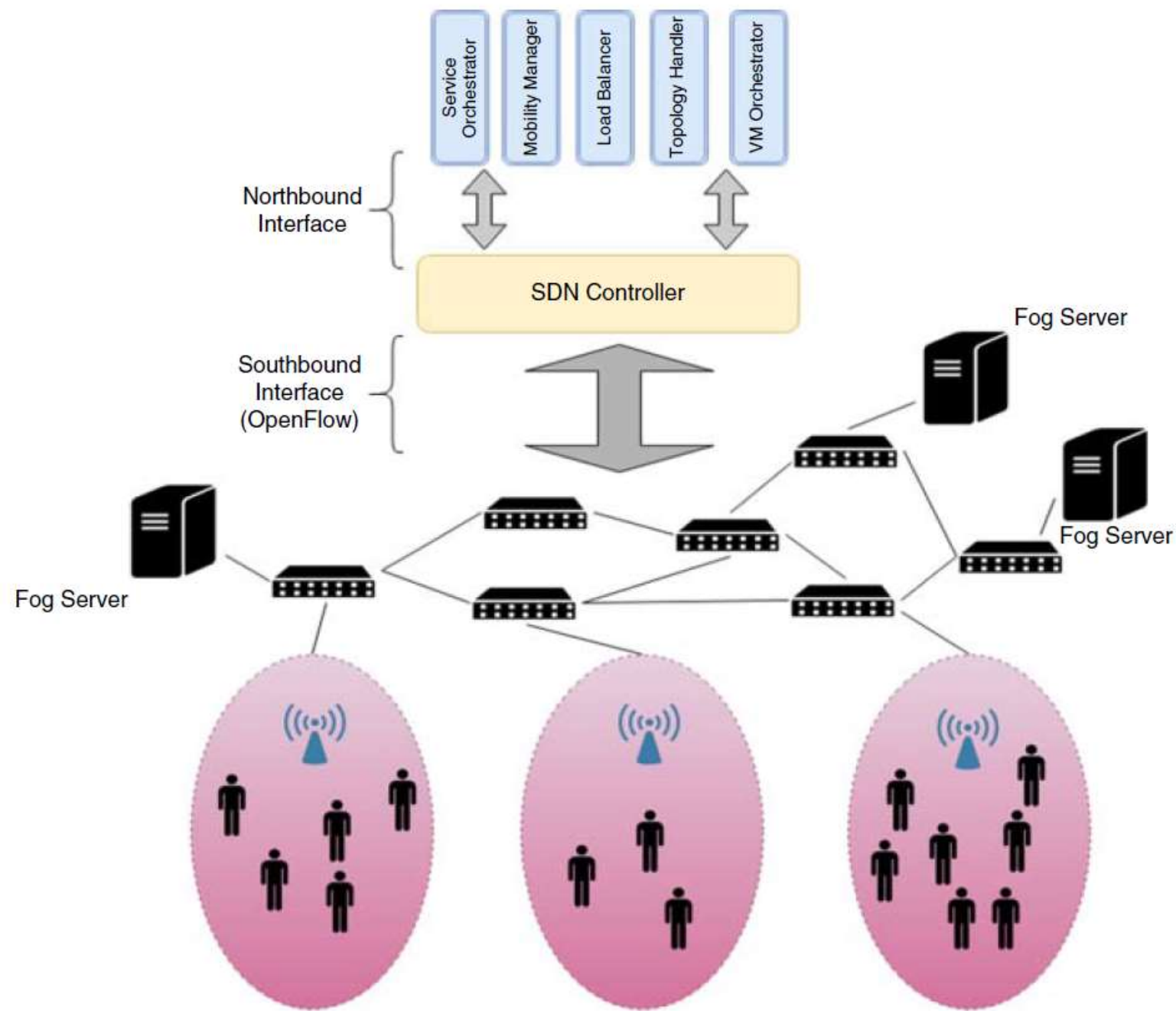


Figure 2.2 Fog computing with SDN as the network orchestrator.

Networking Challenges

- **Centralized vs. Distributed Control Plane**

- **Need for Distributed Control Plane**

- In a federated edge environment, a single control plane may not suffice due to the scale and diversity of resources.
 - A distributed control plane allows for localized management while maintaining global synchronization.

- **Inter-Domain Communication**

- Establishing reliable communication channels between different controllers to ensure consistent service delivery and resource allocation.

Networking Challenges

4. Future Research Directions

- **Implementation of Wireless Networking and SDN**

- Exploring how SDN can be effectively applied in wireless networks to support mobile edge computing.

- **Standardization of Interfaces for Interoperability**

- Developing standards for northbound interfaces to facilitate communication between applications and SDN controllers.

- **Enhancing Programmability of Existing Standards**

- Researching ways to improve the programmability of SDN protocols to support more complex edge computing scenarios.

- **Investigating Scalability of SDN Planes**

- Addressing the challenges of scaling SDN control planes to manage a wide geographic distribution of edge resources.

II. The Management Challenge in Federating Edge Resources

1. Introduction to Management Challenges

- **Overview of Edge Computing Management**

- Edge computing introduces a **layer of complexity** between cloud data centers and user devices, necessitating effective management strategies.
- The **need for managing multiple edge nodes** across different geographic locations to **create a cohesive global architecture**.

2. Key Management Challenges

- **Discovery of Edge Nodes**

- **Importance of Discovery**

- **Efficient resource selection** is critical when **edge nodes** are geographically dispersed and loosely coupled.
 - **Discovery mechanisms** must ensure that edge nodes are **visible to both applications and cloud servers**.

- **Challenges**

- Determining the best practices for discovery: self-initiated, externally monitored, or a hybrid approach.

- **Proposed Solutions**

- Implementing lightweight protocols and handshaking mechanisms to facilitate discovery

Table 2.2 Management challenges, the need for addressing them, and potential solutions in federating edge resources.

Management challenge	Why should it be addressed?	What is required?
Discovery of edge nodes	To select resources when they are geographically spread and loosely coupled	Lightweight protocols and handshaking
Deployment of service and applications	Provide isolation for multiple services and applications	Monitoring and benchmarking mechanisms in real-time
Migrating services	User mobility, workload balancing	Low overhead virtualization
Load balancing	To avoid heavy subscription on individual nodes	Auto-scaling mechanisms

II. The Management Challenge in Federating Edge Resources

- **Deploying Services and Applications**

- **Challenges in Deployment**

- Services must be offloaded to edge nodes based on their capabilities and the requirements of the applications.
 - Real-time benchmarking of multiple edge nodes is essential to meet service objectives.

- **Proposed Solutions**

- Development of monitoring and benchmarking mechanisms to assess edge node capabilities in real-time.

- **Migrating Services Across the Edge**

- **Challenges of Migration**

- Existing technologies (VMs, containers) face challenges in real-time migration due to resource constraints at the edge.
 - The need to consider the shortest path for migrating services between edge nodes.

- **Proposed Solutions**

- Low-overhead virtualization techniques to facilitate efficient service migration while maintaining performance.

II. The Management Challenge in Federating Edge Resources

- **Load Balancing**

- **Importance of Load Balancing**

- Heavy subscription to certain services can lead to resource allocation issues on edge nodes.
 - Effective load balancing is necessary to ensure fair resource distribution among services.

- **Challenges**

- Traditional load balancing methods may not be suitable due to resource constraints on edge nodes.

- **Proposed Solutions**

- Implementing auto-scaling mechanisms to dynamically allocate resources based on service demand.

II. The Management Challenge in Federating Edge Resources

3. Current Research in Management Challenges

- **Discovery of Edge Nodes**

- Existing techniques vary based on multi-tenant environments (e.g., FocusStack for single-tenant, ParaDrop and EaaS for multi-tenant).
- Need for further research to enable discovery in federated edge environments.

- **Deploying Services**

- Focus on pre-deployment resource provisioning to match application requirements with available resources.
- Post-deployment techniques are crucial for managing variability in workloads on edge nodes.

- **Migrating Services**

- Current strategies for live migration of VMs are time-consuming and may not be suitable for real-time applications.
- Research into lightweight technologies (e.g., containers) for faster migration is necessary.

II. The Management Challenge in Federating Edge Resources

- **Load Balancing**

- Existing monitoring systems for distributed environments often do not scale well or are resource-intensive.
- Need for lightweight monitoring and auto-scaling methods tailored for edge environments.

4. Addressing the Management Challenges

- **Edge-as-a-Service (EaaS) Platform**

- **Overview**

- Targets the discovery challenge with a lightweight discovery protocol for homogeneous edge resources.
- Operates in a three-tier environment (cloud, edge nodes, user devices).

- **Functionality**

- Master node communicates with edge nodes to manage resources and monitor performance.
- Tested in real-world scenarios (e.g., online gaming) to improve application performance.

II. The Management Challenge in Federating Edge Resources

- **Edge Node Resource Management (ENORM) Framework**

- **Overview**

- Focuses on deployment and load balancing challenges without a centralized master node.
 - Allows cloud servers to offload workloads to edge nodes while ensuring QoS.

- **Functionality**

- Implements a provisioning mechanism for deploying workloads and monitoring performance.
 - Auto-scaling algorithms adjust resources based on application priority and performance metrics.

II. The Management Challenge in Federating Edge Resources

5. Future Research Directions

- **Coordinating Management Tasks**

- Need for protocols to **manage heterogeneous edge nodes** across multiple collections.

- **Real-Time Benchmarking Services**

- Development of mechanisms for **reliable benchmarking of edge nodes** to meet service-level objectives.

- **Facilitating Rapid Migration**

- Research into **alternate virtualization technologies** for faster service **migration** across edge resources.

- **Fine-Grain Resource Allocation**

- Investigating methods for **dynamic resource allocation** that meet specific application requirements without over-provisioning.

III. Miscellaneous Challenges in Federating Edge Resources

1. Introduction to Miscellaneous Challenges

- **Overview of Edge Computing**

- Edge computing aims to bring computational resources closer to end-users, reducing latency and improving service quality.
- While networking and management challenges are critical, additional miscellaneous challenges must also be addressed for effective federation of edge resources.

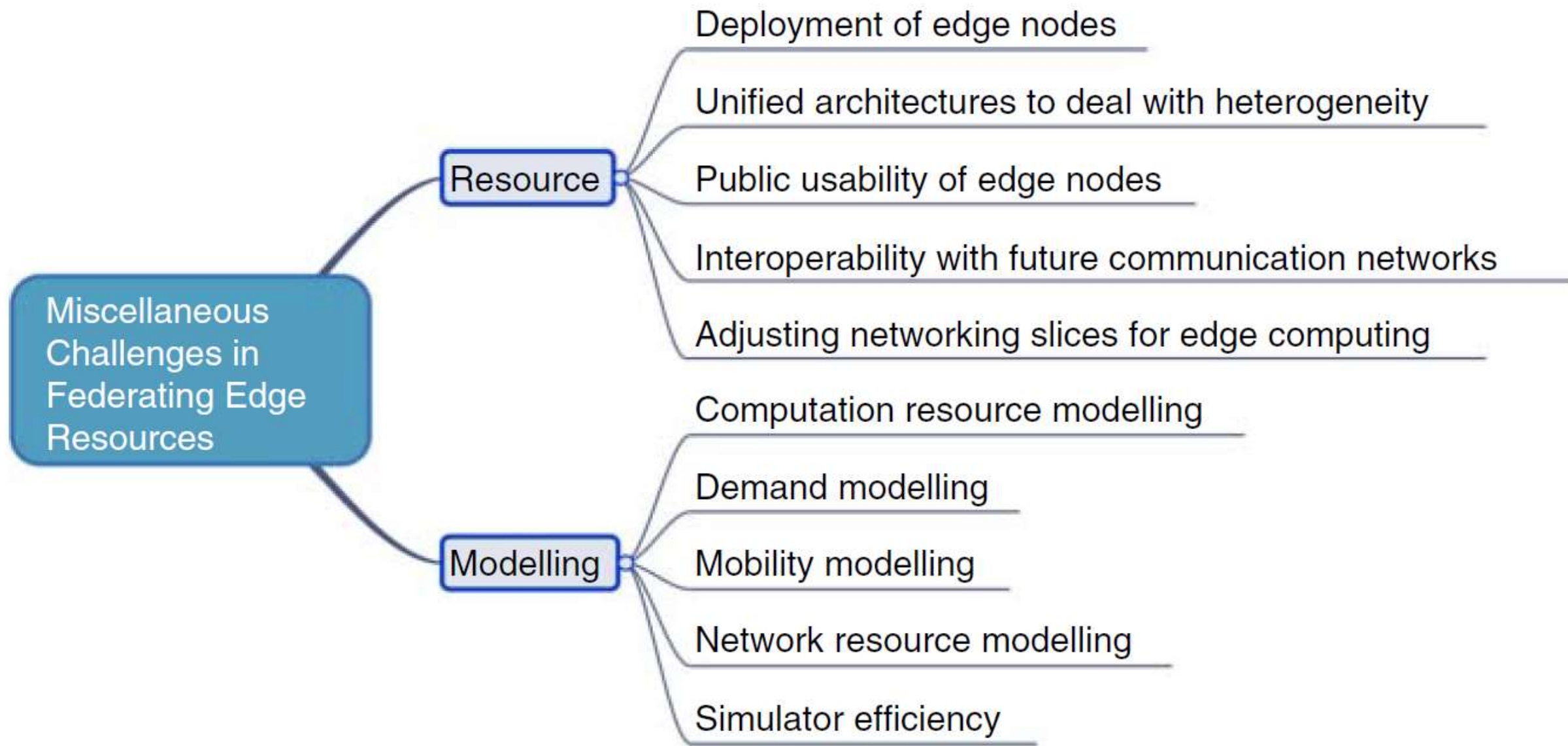


Figure 2.3 Resource and modeling challenges in federating edge resources.

III. Miscellaneous Challenges in Federating Edge Resources

2. Resource Challenges

- **Defined Edge Nodes**

- **Challenge Overview**

- **Uncertainty regarding the types of edge nodes:** traffic routing nodes (routers, switches) vs. dedicated computing nodes (micro clouds).
 - The need to **determine how these nodes will coexist and function together.**

- **Implications**

- The transition to **edge-enabled traffic routing nodes** may require **significant upgrades to existing** infrastructure.
 - Understanding the **business use cases for different types of edge nodes** is essential for effective deployment.

III. Miscellaneous Challenges in Federating Edge Resources

- **Unified Architectures to Account for Heterogeneity**

- **Challenge Overview**

- The **diversity of edge nodes** (varying performance and computational capabilities) **complicates the creation of a coherent architecture**.
 - Need for **unified interoperable standards** to facilitate communication and resource sharing among **heterogeneous edge devices**.

- **Implications**

- **Current virtualization and containerization solutions may not be suitable** for all hardware architectures.
 - **Developing a unified architecture is crucial for seamless integration and operation of edge resources.**

III. Miscellaneous Challenges in Federating Edge Resources

- **Public Usability of Edge Nodes**

- **Challenge Overview**

- Ensuring that **edge nodes are accessible for computation and data processing** from user devices and sensors.
 - **Concerns regarding auditing, accessibility interfaces, billing models, and security measures** for public edge nodes.

- **Implications**

- Addressing these concerns is vital for the widespread adoption and usability of edge computing solutions.

III. Miscellaneous Challenges in Federating Edge Resources

- Interoperability with Communication Networks

- Challenge Overview

- The network itself is a critical resource that influences the performance of edge computing systems.
 - Need for resource management strategies that consider both computational and network resources.

- Implications

- The emergence of 5G and MEC (Multi-access Edge Computing) standards presents opportunities and challenges for edge computing integration.
 - Understanding how edge services will interact with future communication networks is essential for effective deployment.

III. Miscellaneous Challenges in Federating Edge Resources

- Network Slices for Edge Systems

- Challenge Overview

- Network slicing allows for the creation of logical networks tailored to specific services or groups of services.
 - The challenge lies in managing and scaling these slices effectively within edge computing environments.

- Implications

- Fine-grained slicing can enhance service quality but introduces complexity in resource allocation and management.

III. Miscellaneous Challenges in Federating Edge Resources

3. Modeling Challenges

- **Computational Resource Modeling**

- **Challenge Overview**

- Edge servers provide **computational power through virtualization techniques** (VMs, containers).
 - A **simulation environment must support** the creation, resizing, and migration of virtual resources.

- **Implications**

- **Accurate modeling of CPU, memory, and network resource consumption** is necessary for effective edge resource management.

III. Miscellaneous Challenges in Federating Edge Resources

- Demand Modeling

- Challenge Overview

- Modeling the load on edge computing systems requires understanding user behavior and application traffic patterns.
 - The complexity arises from the heterogeneity of devices and applications generating traffic.

- Implications

- Developing profiles for users and applications with predefined distributions can aid in demand modeling.

III. Miscellaneous Challenges in Federating Edge Resources

- **Mobility Modeling**

- **Challenge Overview**

- Mobility impacts **service delivery and resource allocation** in edge computing environments.
 - The simulation environment must **capture various forms of mobility** (e.g., users moving between edge servers).

- **Implications**

- **Realistic mobility modeling is essential** for designing experiments and understanding the dynamics of edge computing.

III. Miscellaneous Challenges in Federating Edge Resources

- Network Modeling

- Challenge Overview

- Accurate modeling of network performance and behavior is critical for edge system operation.
 - Dynamic workloads and varying access technologies (Wi-Fi, cellular) complicate network delay modeling.

- Implications

- An edge simulation tool must scale network resources rapidly to accommodate multiple network slices.

III. Miscellaneous Challenges in Federating Edge Resources

- **Simulator Efficiency**

- **Challenge Overview**

- **Simulators must be scalable, extensible, and user-friendly** to model federated edge resources effectively.
 - The time complexity of simulators should account for a large number of devices and users.

- **Implications**

- Developing efficient simulators is crucial for conducting repeatable and scalable experiments in edge computing.